Comparing Futures and Forwards

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Let the constant *daily* risk-free rate be r, so the daily gross risk-free factor is $R = 1 + r$.
For a given maturity n:

- F_t = futures price (for maturity n) at day t.
- X_t = number of futures contracts held over $[t, t+1]$.
- F_0^{fwd} = current forward price for maturity n
- Y_t = number of forward contracts held

Part A: Daily pricing

A.1 One day until expiration

One day left (from $n-1$ to n).
We hold X futures contracts, and the futures price moves from F_{n-1} to F_n.

$$\boxed{\text{Daily gain/loss on expiration } = X\,(F_n - F_{n-1}).}$$

This gain is realized *at* expiration. No compounding needed.

A.2 Two days until expiration

Two days left (from $n-2$ to n).
At day $n-2$ we hold X_{n-2} futures contracts, and the futures price is F_{n-2}.
The price at day $n-1$ is F_{n-1}.

- Gain/loss realized at settlement on day $n-1$:

$$\boxed{\text{Gain at } n-1 \; = \; X_{n-2}\,(F_{n-1} - F_{n-2}).}$$

- Compounded value of this gain at expiration:

$$\boxed{\text{Value at } n \text{ of this gain } = X_{n-2}\,(F_{n-1} - F_{n-2})\,R.}$$

A.3 General day t:

Fix any $t \in \{0, 1, \ldots, n-1\}$.

- We hold X_t contracts over $[t, t+1]$.
- The futures price moves from F_t to F_{t+1}.
- Gain at $t+1 \;=\; X_t\,(F_{t+1} - F_t)$.
- Value at n of the gain from day $t \;=\; X_t\,(F_{t+1} - F_t)\,R^{n-t-1}$.

The *total* futures P&L at expiration: $\boxed{V_n^{\text{fut}} \;=\; \sum_{t=0}^{n-1} X_t\,(F_{t+1} - F_t)\,R^{n-t-1}.}$

A.4 Forward + bond portfolio and choice of Y_0

n days until expiration.
The current forward price is F_0^{fwd}.
Consider the portfolio:

- Invest the dollar amount F_0^{fwd} in the risk-free bond at time 0.
- Take a long forward position of size Y_0 on the same underlying, with delivery price $K = F_0^{\text{fwd}}$ and maturity at day n.

At expiration:

- The bond grows to $F_0^{\text{fwd}}\,R^n$.
- The long forward position of size Y_0 pays $Y_0\,(S_n - F_0^{\text{fwd}})$, where S_n is the spot price at expiration. Using convergence, $S_n = F_n$.

Total terminal value:

$$\begin{aligned}
V_n^{\text{fwd+bond}} &= F_0^{\text{fwd}}\,R^n + Y_0\,(F_n - F_0^{\text{fwd}}) \\
&= F_0^{\text{fwd}}\,(R^n - Y_0) + Y_0 F_n.
\end{aligned}$$

We want this terminal value to be *independent* of the initial forward price F_0^{fwd}. This happens if the coefficient of F_0^{fwd} is zero: $R^n - Y_0 = 0 \quad\Longrightarrow\quad \boxed{Y_0 = R^n.}$

$$\boxed{V_n^{\text{fwd+bond}} = R^n F_n,}$$

which depends only on the terminal underlying price and not on F_0^{fwd}.

A.5 Choosing Y_0 and $(X_0, \ldots, X_{n-1})$

The futures-only terminal value: $V_n^{\text{fut}} = \sum_{t=0}^{n-1} X_t \, (F_{t+1} - F_t) \, R^{\,n-t-1}$.
We want a choice of futures positions (X_t) that replicates the payoff of a forward with notional Y_0 and delivery price F_0: Forward payoff at $n = Y_0 \, (F_n - F_0)$.
Define

$$\boxed{X_t \;=\; \frac{Y_0}{R^{\,n-t-1}}, \qquad t = 0, 1, \ldots, n-1.}$$

Substituting into V_n^{fut}:

$$\begin{aligned} V_n^{\text{fut}} &= \sum_{t=0}^{n-1} \left(\frac{Y_0}{R^{\,n-t-1}} \right) (F_{t+1} - F_t) \, R^{\,n-t-1} \\ &= Y_0 \sum_{t=0}^{n-1} (F_{t+1} - F_t) \\ &= Y_0 \, (F_n - F_0), \end{aligned}$$

which *exactly* matches the forward payoff $Y_0(F_n - F_0)$ for any price path $\{F_t\}$.
The two portfolios:

- **Forward + bond**:
$$V_n^{\text{fwd+bond}} = F_0^{\text{fwd}} \, R^n + Y_0 \, (F_n - F_0^{\text{fwd}}).$$
- **Futures + bond**:
$$V_n^{\text{fut+bond}} = F_0^{\text{fwd}} \, R^n + Y_0 \, (F_n - F_0).$$

For these terminal values to agree for all possible F_n we need

$$F_0^{\text{fwd}} \, R^n + Y_0(F_n - F_0^{\text{fwd}}) = F_0^{\text{fwd}} \, R^n + Y_0(F_n - F_0),$$

$$\implies Y_0(F_n - F_0^{\text{fwd}}) = Y_0(F_n - F_0) \quad \implies \quad \boxed{F_0^{\text{fwd}} = F_0} \quad \text{(assuming } Y_0 \neq 0\text{)}.$$

With a deterministic risk-free rate, the forward and futures prices for the same underlying and maturity must be equal at time 0, and by taking $X_t = \frac{Y_0}{R^{\,n-t-1}}$ we can use futures to replicate the payoff of a forward of size Y_0.
If we use $Y_0 = R^n$, then $X_t = \frac{R^n}{R^{\,n-t-1}} = R^{\,t+1}$, and both the forward+bond portfolio and the futures+bond portfolio have the common terminal value $R^n F_n$.

Part B: Portfolio value and dynamic rebalancing

B.1 Futures: value and rebalancing

Let F_{t_1} be the futures price at time t_1, and F_{t_2} at $t_2 > t_1$.

- We hold X_{t_1} futures contracts over $[t_1, t_2]$.
- Cumulative mark-to-market gain/loss credited to the margin account:

$$\boxed{\Delta V_{t_1 \to t_2}^{\text{fut}} = X_{t_1} \left(F_{t_2} - F_{t_1} \right).}$$

- Immediately after settlement at t_2, the *contract value* of the futures position is zero
- Rebalancing from X_{t_1} to X_{t_2} at time t_2 does not produce additional cash flow.

$$\boxed{V_{t_2}^{\text{fut}} - V_{t_1}^{\text{fut}} = X_{t_1}\left(F_{t_2} - F_{t_1}\right),}$$

where V_t^{fut} is the value of the margin account associated with the futures position.

B.2 Forwards: value and rebalancing

Consider a forward contract on the same underlying with maturity at day n. At time t_1, we enter Y_{t_1} forward contracts with delivery price $K = F_{t_1}^{\text{fwd}}$ so that the position has zero value at t_1. Let F_t^{fwd} be the forward price for maturity n at time t, and $B(t,n) := \frac{1}{R^{n-t}} = (1+r)^{-(n-t)}$ be the discount factor from n back to t.

Value at time $t_2 > t_1$:

$$\boxed{V_{t_2}^{\text{fwd}} = Y_{t_1}\left(F_{t_2}^{\text{fwd}} - K\right) B(t_2, n).}$$

Since $K = F_{t_1}^{\text{fwd}}$,

$$\boxed{V_{t_2}^{\text{fwd}} = Y_{t_1}\left(F_{t_2}^{\text{fwd}} - F_{t_1}^{\text{fwd}}\right) B(t_2, n).}$$

Rebalancing at t_2.

- We close the old position Y_{t_1} (by entering $-Y_{t_1}$ forwards at strike $F_{t_2}^{\text{fwd}}$), realizing a cash gain/loss exactly equal to $V_{t_2}^{\text{fwd}}$ above.
- We open a new forward position of size Y_{t_2} at the current forward price $F_{t_2}^{\text{fwd}}$, which has zero value at initiation.

Change in portfolio value:

$$\boxed{V_{t_2}^{\text{fwd}} - V_{t_1}^{\text{fwd}} = Y_{t_1}\left(F_{t_2}^{\text{fwd}} - F_{t_1}^{\text{fwd}}\right) B(t_2, n),}$$

$V_{t_1}^{\text{fwd}} = 0$ at inception.

B.3 Continuous-time limit and forward vs futures hedge size

Assume for the same maturity n, the forward and futures prices coincide: $F_t^{\text{fwd}} = F_t$.

$$B(t_2, n) \approx B(t_1, n) =: B(t, n)$$

For small Δt, and F^{fwd} and F share the same increment $\Delta F_t = F_{t_2} - F_{t_1}$.

$$\Delta V_t^{\text{fut}} \approx X_t\, \Delta F_t,$$
$$\Delta V_t^{\text{fwd}} \approx Y_t\, B(t,n)\, \Delta F_t,$$

Limit $\Delta t \to 0$:

$$\boxed{dV_t^{\text{fut}} = X_t\, dF_t, \qquad dV_t^{\text{fwd}} = Y_t\, B(t,n)\, dF_t,}$$

For the *same* instantaneous sensitivity: $dV_t^{\text{fut}} = dV_t^{\text{fwd}} \quad \Longrightarrow \quad X_t\, dF_t = Y_t\, B(t,n)\, dF_t,$

$$\boxed{X_t = Y_t\, B(t,n) \quad \Longleftrightarrow \quad Y_t = \frac{X_t}{B(t,n)} = X_t\,(1+r)^{\,n-t}.}$$

$B(t,n) < 1$ when $r > 0 \implies |Y_t| > |X_t| \quad \text{for } t < n.$
If a given risk exposure is hedged using 100 futures contracts, the *equivalent* forward hedge must satisfy $Y_t = \frac{100}{B(t,n)} = 100\,(1+r)^{\,n-t} > 100,$, for positive r.

$$\boxed{\text{The corresponding forward position must be } \textit{larger} \text{ than 100 contracts.}}$$

The forward position is less sensitive to F_t at time t, so it must be scaled up relative to a futures position to provide the same hedge.

AI usage: ChatGPT was used for copy-editing of the homework